

MR Flow Quantification Sensor at Catheter Scale

Executive Technology Brief

Domain: Magnetic Resonance (MRI/NMR) + Flow Quantification + Miniaturization

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: CSF flow measurement requires clinical MRI (expensive, not continuous)

Mechanism: Miniaturized permanent magnet + RF coil + gradient coil enables phase-contrast MRI flow measurement at catheter scale

Architecture: Implantable MR flow sensor that quantifies CSF flow rate in shunt catheter using phase-contrast MRI principles. Miniaturized magnet + RF coil integrated into catheter or external patch.

Governing Model: Phase-contrast MRI flow measurement: bipolar gradient encodes velocity as phase shift; two acquisitions with opposite gradient polarities yield phase difference proportional to velocity.

Key Equations:

$\omega = \gamma * B_0$ [Larmor frequency for P-29 miniaturized magnet at catheter scale]

$d\phi = \gamma * M_1 * v$ [phase encoding of CSF flow velocity in P-29 catheter-scale MR sensor]

$VENC = \pi / (\gamma * M_1)$ [velocity encoding]

$Q = v * A$ [volumetric flow from velocity]

$SNR \sim B_0 * \sqrt{N * voxel_volume} / \sqrt{BW}$ [SNR scaling]

Kill Condition: Miniaturization to catheter scale cannot achieve $SNR > 10:1$ at any B_0 field (physically blocked)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.